DEPARTMENT: Information Science and Engineering			
Course Code: 24CB410	Course Title : OPERATING SYSTEMS		
Total Credits: 04 L-45: T-00: P-45: SL and TW-30	SDGs addressed: 4	Course Category: PCC	Course Type: Integrated

Teaching Learning Process		Total Contact Hours /semester	Assessment Weightage and Marks			
	Lecture	45		CIE	SEE	Total
	Tutorial	-	Weightage, %	40	60	100
	Practical	30	Maximum Marks	40	60	100
	Self-learning and Teamwork	45	Minimum Marks	20	25	45*
	Total	120				

Note: *For passing the student must score a Minimum of 45 Marks (CIE+SEE: 20 + 25 or 21 + 24)

COURSE PRE-REQUISITE: Boolean algebra and knowledge of programming.

COURSE OBJECTIVE: To understand the fundamental concepts of Operating Systems.

COURSE OUTCOMES (COs)

By the end of the course, students will be able to:

CO	Course Outcomes	Highest Level of Cognitive Domain	Working Knowledge
CO1	Describe the basic structure, functions and services of operating systems.	L2	WK3 WK6
CO2	Illustrate the performances of different scheduling algorithms and compare them.	L3	WK3 WK6
CO3	Describe the different processes synchronization and deadlock handling techniques.	L3	WK3 WK6
CO4	Demonstrate how memory virtualization addresses limitations of physical memory.	L3	WK3 WK6
CO5	Demonstrate the structure and organization of file systems through practical scenarios.	L3	WK3 WK6

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 - Create

Course Syllabus:

Unit No.	Content	Lecture Hours	Tutorial
1	**Introduction:** Computer system organization, computer system architecture, OS structure, OS operation, process management, memory management, storage management, computing environment, OS services, user OS interpreter, system calls, types of system call, virtual machine. **Self-Learning and Termwork:** Analyze various operating system structures and environments.	09	-
2	**Process Concept:** process models, process scheduling, operation on processes, Inter Process Communication, communication in client server systems. Threads: thread states, types of threads, thread priority, multithreading **Process management:** Scheduling: short-term scheduler types of schedulers, non-preemptive & pre-emptive strategies, scheduling algorithms. **Self-Learning and Termwork:** Implement hybrid scheduling algorithms, understand practically multi-core and multi-threading models.	09	-
3	**Synchronization:** concepts, critical section problem, Peterson's solution, synchronization hardware, semaphores, bounded buffer problem, readers-writer's problem, dining philosopher's problem **Deadlocks:** deadlocks & starvation, conditions for deadlocks, deadlock detection, deadlock prevention, deadlock recovery, deadlock avoidance, starvation. **Self-Learning and Termwork:** Analize various synchronization techniques.	09	-
4	**Memory management:** Key Characteristics, memory management functions, logical address & physical address space, swapping, contiguous memory allocation, paging, segmentation Virtual memory management: overview demand paging, page replacement. **Self-Learning and Termwork:** Implement continuous memory allocation. Implement analyze belady's anomaly.	09	-
5	**Device & file management:** Device Characteristics, types, device controller, operations, buffering & types of buffering, clock, disk, management, disk arm scheduling policies, RAID, File system, server, management, design, organization, directory, sharing, blocking, management, allocation, file system reliability. **Self-Learning and Termwork:** Implement RAID, apply operating system concepts to develop real-time application.	09	-

Laboratory

Week	List of Experiments/ Programs	No. of Hours
1	Implementation of Basic Linux commands	2
2	Implementation of different commands for shell programming (output, input, exit status, arithmetic operations)	2
3	Implementation of process related system calls	2
4	Implement the concept of Dining Philosopher's problem	2
5	Implementation of first come first serve.	2
6	Implementation of shortest job scheduling algorithm	2
7	Implementation of Priority scheduling algorithms	2
8	Implementation of Round Robin CPU scheduling algorithms	2
9	Implementation of Banker's algorithm for Deadlock Avoidance	2
10	Implementation of Banker's algorithm for Deadlock Detection	2

11	Implementation of page replacement algorithms	**2**
12	Implementation of Memory Management using Address Translation	**2**
13	Implement Paging technique for memory management	**2**
14	Revision	**2**
15	Laboratory Test	**2**

Text Books:

1. Operating System Principles, Abraham Silberschatz, Peter Galvin, Greg Gagne, Wiley.
2. Operating Systems, A concept–based evolutionary approach, P Chakraborty, Jaico publishing house.

Reference Books

1. Modern Operating Systems, Andrew S Tanenbaum, Herbert Bos, 4th Edition, Pearson Edition.

Journals/Magazines:

1. journal-of-parallel-and-distributed-computing https://www.sciencedirect.com/journal/journal-of-parallel-and-distributed-computing

Web/Digital resources:

1. https://ocw.mit.edu/courses/6-828-operating-system-engineering-fall-2012/pages/lecture-notes-and-readings/

SWAYAM/NPTEL:

1. Operating System Fundamentals, IIT Kharagpur https://nptel.ac.in/courses/106105214
2. Introduction to Operating Systems, IIT Madras https://nptel.ac.in/courses/106106144

Course Articulation Matrix:

COURSE OUTCOMES ↓	PROGRAM OUTCOMES											PROGRAM SPECIFIC OUTCOMES		
	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PSO1**	**PSO2**	**PSO3**
CO1	3	3	3	3								3	3	
CO2	3	3	3	3	2							3	3	
CO3	3	3	3	3	2							3	3	
CO4	3	3	3	3	2							3	3	
CO5	3	3	3	3	2							3	3	

High – 3, Medium – 2, Low – 1